

KrishiAI – Product & Business Requirements Document

■ Product Requirements Document (PRD)

1. Project Title: KrishiAI – Smart Farming Solution

2. Objective:

To revolutionize agriculture using AI-powered tools that help farmers plan crops, detect diseases, manage farms, and make data-driven decisions for higher yield and sustainability.

3. Product Overview:

KrishiAI is an intelligent platform combining AI, data analytics, IoT, and voice AI to provide crop planning, disease detection, farm management, and analytics. The platform includes web and mobile applications for farmers to access insights in real-time.

4. Target Users:

- Small and large-scale farmers
- Agricultural consultants
- Agri-tech companies and cooperatives

5. Key Features:

- AI Crop Planning – suggests crops based on soil, weather, and demand.
- Disease Detection – identifies plant issues using AI image analysis.
- Farm Management – tracks farming tasks and resources.
- AI Farmer Assistant – provides 24/7 support via chat/voice.
- Analytics Dashboard – visualizes performance data.
- Offline Access – mobile app works without internet.
- Voice Marketplace (FreshFarm) – enables voice-based ordering.

6. Functional Requirements:

- 1 User registration and login with verification.
- 2 Farm profile setup with soil, crop, and location details.
- 3 AI modules for prediction and disease detection.
- 4 Dashboard with analytics and visualization.
- 5 Voice interaction for assistance and ordering.
- 6 Syncing between web and mobile platforms.
- 7 Notification and alert system for farmers.

7. Non-Functional Requirements:

- Performance: Handle 10,000+ users simultaneously.
- Scalability: Easily expandable on cloud infrastructure.
- Usability: Designed for non-technical farmers.
- Reliability: Offline capability and auto backup.

- Security: End-to-end data encryption.

8. Technology Stack:

- Frontend: ReactJS (Web), React Native (Mobile).
- Backend: Python (Flask/Django).
- AI Models: TensorFlow, OpenCV.
- Database: Firebase / MongoDB.
- Cloud: AWS / Google Cloud.
- Voice AI: NLP + Speech Recognition APIs.

9. Deliverables:

- Web dashboard for AI farming insights.
- Mobile application for on-field access.
- Voice AI marketplace (FreshFarm).
- Admin panel for monitoring and analytics.

■ Business Requirements Document (BRD)

1. Purpose:

To develop a unified AI platform that empowers farmers to improve yield, reduce crop loss, and adopt smart agriculture practices through automation and analytics.

2. Business Goals:

- 1 Increase farmer productivity by 30–40%.
- 2 Reduce operational costs using AI optimization.
- 3 Promote digital agriculture and sustainability.
- 4 Connect farmers directly to buyers.
- 5 Expand user base to 50,000+ within one year.

3. Stakeholders:

- Farmers – end users.
- KrishiAI Team – developers and data scientists.
- Government/NGOs – digital agriculture partners.
- Agri-businesses – marketplace partners.

4. Business Benefits:

- Boosts farm productivity and profitability.
- Reduces crop loss through early disease detection.
- Builds agricultural data ecosystem.
- Encourages sustainable and smart farming practices.

5. Success Metrics:

- Active Users: 10,000+ farmers.

- Yield Improvement: 30–40%.
- Cost Reduction: 20%.
- AI Accuracy: 90–95% for disease detection.
- System Uptime: 99.5%.

6. Risks & Mitigation:

- Low adoption → Offer free demos and training sessions.
- Internet issues → Provide offline functionality.
- Data security → Use encrypted cloud storage.
- AI model errors → Regular retraining with updated datasets.

7. Revenue Model:

- Free tier for basic tools.
- Premium subscription for advanced features.
- Commission from FreshFarm marketplace transactions.
- API licensing for third-party integrations.

8. Future Enhancements:

- Integration with IoT soil and weather sensors.
- Drone-based crop analysis.
- Blockchain-based crop traceability.
- Multilingual voice AI support for regional farmers.

Conclusion:

KrishiAI is a comprehensive digital agriculture platform that leverages AI and analytics to make farming more sustainable, efficient, and profitable. With continuous innovation and farmer-focused features, it aims to lead the next wave of smart farming transformation.